

Q1)

a) $q_0 _$, $0q_1 _$, $01q_2 _$, $0q_211 _$, $q_2001 _$
 $1q_101 _$, $11q_01 _$, $110q_1 _$, $1101q_2 _$, $110q_211 _$
 $11q_2001 _$, $111q_101 _$, $1111q_01 _$, $11110q_1 _$, $111101q_2 _$

b) The largest number of characters by which any two neighbouring configurations C_i and C_{i+1} can differ for this TM is 3

Why? Firstly, in this TM M_2 , we will not write or override some character to empty string, so we don't consider to shift part of the configuration.

From C_i to C_{i+1} , consider two cases: encoded TM state move right or left.

Difference between C_i and C_{i+1}

Right: only move

2

overwrite the character

2

left: only move { if at head of string

0

otherwise

2

overwrite the character { if at head of string

1

otherwise

3

Overall, the largest difference is 3.

$$Q2) \overline{L_2} = \{ \langle M \rangle \mid \forall n \in \mathbb{N}, 0^n 1^n \in L(M) \}$$

Proof $\text{HALT} \leq_m \overline{L_2}$

Consider TM P on input $\langle M, w \rangle$:

1. Construct the TM M_0 as follows:

$M_0 =$ on input x

1. Pass
2. Run M on w
3. Accept

2. Return $\langle M_0 \rangle$

Proof that $\langle M_0 \rangle \in \overline{L_2}$ iff $\langle M, w \rangle \in \text{HALT}$

($\Leftarrow$) Suppose that $\langle M, w \rangle \in \text{HALT}$,
 then for every input x , M_0 reaches line 2.
 when it does, M will eventually halt and M_0 will reach
 line 3. When it does, it will accept every string.
 which of course, it will accept every string in the format
 of $0^n 1^n$ for all $n \in \mathbb{N}$.
 $\Rightarrow \langle M_0 \rangle \in \overline{L_2}$

($\Rightarrow$) Suppose that $\langle M, w \rangle \notin \text{HALT}$
 then for every input x , M_0 reaches line 2.
 when it does, M will loop and so M_0 will never reach
 line 3 and accept. So M_0 will not accept any input.
 $\Rightarrow \langle M_0 \rangle \notin \overline{L_2}$

Since we know HALT is not co-recognizable, then $\overline{L_2}$ is not
 co-recognizable. And since $\overline{L_2}$ is the complement of L_2 ,
 $\therefore \overline{L_2}$ is not co-recognizable, then L_2 is not recognizable.

Q3: Proof $\text{HALT} \leq_m L_3$

Consider the TM P "On input $\langle M, w \rangle$:

1. Construct the TM M_0 as follows :

$M_0 =$ "On input x :

1. Pass

2. Run M on w .

3. if $x == 01$, accept , otherwise reject

2. Return $\langle M_0 \rangle$."

Pf that $\langle M_0 \rangle \in L_3 \iff \langle M, w \rangle \in \text{HALT}$.

($\Leftarrow$) Suppose that $\langle M, w \rangle \in \text{HALT}$.

1. Then , for every input x , M_0 reaches line 2.

2. When it does , M will eventually halt and M_0 will reach line 3.

3. When it does , it will accept iff $x == 01$

$\Rightarrow L(M_0) = \{0^n 1^n \mid \exists n \in \mathbb{N}\}$

$\Rightarrow \langle M_0 \rangle \in L_3$

($\Rightarrow$) Suppose that $\langle M, w \rangle \notin \text{HALT}$.

1. Then , for every input x , M_0 reached line 2.

2. When it does , M will loop , and so M_0 will never reach line 3 and accept

3. So M_0 will not accept any input

$\Rightarrow L(M_0) \neq \{0^n 1^n \mid \exists n \in \mathbb{N}\}$

$\Rightarrow \langle M_0 \rangle \notin L_3$

$\therefore \text{HALT}$ is not co-recognizable , $\therefore L_3$ is not co-recognizable

Q 4) L_4 is recognizable, but not co-recognizable and is not decidable.

Proof L_4 is recognizable:

Let $w_0, w_1, \dots$ be an effective enumeration of Σ^*

Let $M_{L_4} =$ On input $\langle M \rangle$:

1. For $k=0$ to ∞ :
2. For $i=0$ to k :
3. Run M on w_i for k steps
4. If it accepts, accept.

Argue:

① If $\langle M \rangle \in L_4$:

then there exist at least one string w , $\exists k, \in \mathbb{N}$. M accept w in k steps. So M_{L_4} accept $\langle M \rangle$ in $k=k$ steps

②. If $\langle M \rangle \notin L_4$:

then for any string w , M will not accept w within any k steps, where $k \in \mathbb{N}$. which implies that M_{L_4} does not accept.

To prove L_4 is not co-recognizable

Proof $HALT \leq_m L_4$

• Consider the TM P on input $\langle M, w \rangle$

1. Construct the TM M_0 as follows.

$M_0 =$ "On input x :

1. Pass

2. Run M on w

3. Accept "

2. Return $\langle M_0 \rangle$."

Prove $\langle M_0 \rangle \in L_4$ iff $\langle M, w \rangle \in HALT$

$\Leftarrow$ Suppose that $\langle M, w \rangle \in HALT$

Then, for every input x , M_0 reaches line 2.

When it does, M will eventually halt and M_0 will reach line 3 and accept.

So M_0 will accept every input (and therefore will accept at least one input)

$\Rightarrow \langle M_0 \rangle \in L_4$

$\Rightarrow$ Suppose that $\langle M, w \rangle \notin HALT$

Then, for every input x , M_0 reaches line 2.

When it does, M will loop, and so, M_0 will never reach line 3 and accept. So M_0 will not accept any input.

$\Rightarrow \langle M_0 \rangle \notin L_4$

Q5)

$$L_s = \{ \langle M \rangle \mid \overline{L(M)} \subseteq \{0^*1^n \mid n \in \mathbb{N}\} \}$$

not recognizable, not co-recognizable.

Proof $\overline{\text{HALT}} \leq_m L_s$

• Consider the TM P on input $\langle M, w \rangle$

1. Construct the TM M_0 as follows:

$M_0 =$ "On input x :

1. let $k = 2^* \text{length}(x)$

2. Run M on w for k steps.

3. If it halts and $x \notin \{0^*1^n \mid n \in \mathbb{N}\}$, reject, o/w. Accept.

2. Return $\langle M_0 \rangle$.

Prove $\langle M_0 \rangle \in L_s$ iff $\langle M, w \rangle \in \overline{\text{HALT}}$

($\Leftarrow$) Suppose that $\langle M, w \rangle \in \overline{\text{HALT}}$.

Line 1 will generate a k based on length of x . and M_0 always gets to line 2.

When it does, M not halt within k steps for any k , since it loops.

So M_0 accept every input.

$$\Rightarrow L(M_0) = \{w \mid w \in \Sigma^*\}, \quad \overline{L(M_0)} = \{\emptyset\} \subseteq \{0^*1^n \mid n \in \mathbb{N}\}$$

$$\Rightarrow \langle M_0 \rangle \in L_s$$

($\Rightarrow$) Suppose that $\langle M, w \rangle \notin \overline{\text{HALT}}$.

Line 1 will generate a k based on length of x . and M_0 always gets to line 2.

When it does, M will halt within some k steps, for this k ,

So M_0 will reject some $x \notin \{0^*1^n \mid n \in \mathbb{N}\}$

$$\Rightarrow \overline{L(M_0)} \text{ contains some strings } \notin \{0^*1^n \mid n \in \mathbb{N}\}$$

$$\Rightarrow \text{So } \overline{L(M_0)} \not\subseteq \{0^*1^n \mid n \in \mathbb{N}\}$$

$$\Rightarrow \langle M_0 \rangle \notin L_s$$

$\therefore \overline{\text{HALT}}$ is not recognizable, so neither is L_s

Proof $\text{HALT} \leq_m L_5$:

Consider the TM P : "On input $\langle M, w \rangle$:

1. Construct the TM M_0 as follows:

M_0 = "On input x .

1. Pass

2. Run M on w .

3. if $x \in \{0^*1^n \mid n \in \mathbb{N}\}$, reject, otherwise accept.

Prove that $\langle M_0 \rangle \in L_5$ iff $\langle M, w \rangle \in \text{HALT}$.

($\Leftarrow$) Suppose that $\langle M, w \rangle \in \text{HALT}$

Then for every input x , M_0 reaches line 2.

When it does, M will eventually halt, and M_0 will reach line 3.

When it does, it will accept if $x \notin \{0^*1^n \mid n \in \mathbb{N}\}$

$$\Rightarrow L(M_0) = \{w \in \Sigma^* \mid w \notin \{0^*1^n \mid n \in \mathbb{N}\}\}$$

$$\Rightarrow \overline{L(M_0)} = \{0^*1^n \mid n \in \mathbb{N}\} \subseteq \{0^*1^n \mid n \in \mathbb{N}\}$$

$$\Rightarrow \langle M_0 \rangle \in L_5.$$

($\Rightarrow$) Suppose that $\langle M, w \rangle \notin \text{HALT}$

Then for every input x , M_0 reaches line 2.

When it does, M will loop, and so M_0 will never reach line 3 and accept.

So M_0 will not accept any input.

$$\Rightarrow L(M_0) = \{\emptyset\}$$

$$\Rightarrow \overline{L(M_0)} = \{w \in \Sigma^*\}$$

$$\Rightarrow \overline{L(M_0)} \not\subseteq \{0^*1^n \mid n \in \mathbb{N}\}$$

$$\Rightarrow \langle M_0 \rangle \notin L_5$$

$\therefore \text{HALT}$ is not co-recognizable, so neither is L_5

Q6) Tape description on next page:

Pseudocode.

1. Initialize:

1. Set the initial state q_0 on the state Tape.
2. Set current position $(0,0)$ to the Position tape.
3. Load input string w into String tape.
4. Initial all Neighbor Tapes with ϵ

2. Main loop:

5. while (Current state is not q_A or q_R)
6. - read the state from the State Tape
7. - read the final character from all Neighbor Tapes (if it is ϵ , treat it as $\perp$)
8. - read the current string from String Tape.
9. - Use transition function δ to determine the next state, the character to write or erase, and the position of next cell (x,y) .

10. update {

- Based on δ , write/erase the final character of the string in String Tape.
- Update the Position Tape to the new (x,y)
- update the state Tape to new state.
- Based on the new (x,y) in Position tape, update the content at all neighbor tapes. load ϵ if necessary

11. - check current state in state tape.
If it is q_A , accept, if it is q_R , reject.

1. A state tape that holds the current state of M . ex) q_0
2. A position tape keep track with current position (i, j) in $\mathbb{Z}D$ ex) $\#i\#j\#$
3. Neighbor tapes keep track the 8 separate tape that surround the cell position (i, j) , and initial values are Σ .

ex) Neighbor tapes

{	① : $\#i-1\#j-1\#w_1$	where w_i is string $1 \leq i \leq 8$.
	② : $\#i\#j-1\#w_2$	
	③ : $\#i+1\#j-1\#w_3$	
	④ : $\#i-1\#j\#w_4$	
	⑤ : $\#i+1\#j\#w_5$	
	⑥ : $\#i-1\#j+1\#w_6$	
	⑦ : $\#i\#j+1\#w_7$	
	⑧ : $\#i+1\#j+1\#w_8$	

4. String Tape contains current string w . ex) w

Justification -

Using delimiters and multitapes can ensure the TM M to update the relevant data correctly. Since M and P have the similar transition method, the M and P have the same outcomes of acceptance, rejections and looping. The transition function δ is simulated by reading the required tapes, making decisions based on their contents, and writing outputs accordingly, which mirrors the decision-making process in P .

Q7)

let $N^{\text{HALT}} =$ "On input w :

1. For $k = 0$ to ∞ :
2. For $i = 0$ to k :
3. Run M on w for k steps =
4. if M has printed 'T' and won't print 'F' until halt (Using HALT Oracle)
5. Accept
6. if M has printed 'F' and won't print 'T' until halt (Using HALT Oracle)
7. Reject
8. if neither met
9. loop.

Justification:

N^{HALT} simulates M and monitor the output when halting. If the output only contains 'T' and will not print 'F' in future, verifying by the HALT oracle, N^{HALT} accepts. $\leftarrow$

Similarly, If the output only contains 'F' and will not print 'T' in future, verifying by the HALT oracle, N^{HALT} rejects. $\leftarrow$

If the HALT oracle can't verify either situation, which means M does not stabilize on either 'T' or 'F', N^{HALT} will continue looping and without making final decision (M diverges on w then N^{HALT} loops). $\leftarrow$

By constructing N^{HALT} in this way, we ensure that it behaves exactly as described, leveraging the HALT oracle to determine the eventual behavior of M . This guarantees that N^{HALT} will accept, reject, or loop in accordance with M 's convergence or divergence, thereby simulating the looping TM M using an oracle Turing machine. $\leftarrow$

Q8)

let $M =$ " On input w :

1. For $k = 0$ to ∞ :
2. For $i = 0$ to k :
3. Use the HALT Oracle to determine if N^{HALT} has halted within k steps :
4. if the oracle indicates that N^{HALT} accept w , print 'T' to the output tape and move to right
5. if the oracle indicates that N^{HALT} reject w , print 'F' to the output tape
6. else next looping

Justification :

1. When N^{HALT} accepts w , the HALT oracle will eventually confirm this during one of the looping. Once the oracle ensures the accept state, M will start printing 'T' to the output tape and moves the Output head one step to the right.
2. When N^{HALT} rejects w , the HALT oracle will eventually confirm this during one of the looping. Once the Oracle ensures the reject state, M will start printing 'F' to the Output tape.
3. If both condition failed, the N^{HALT} will loop and M continue simulate N^{HALT} indefinitely without converging to 'T' or 'F', ensuring M diverges on w .